# Simulate The Changes In Equivalent Circuit Of Three Phase Transformer When The System Undergoes Fault And a Phase Is Lost

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## Abstract

The simulation of a three-phase transformer (440V/33 kV, 112 kVA) that experiences a sudden fault that results in the lost of one phase is shown in this study. Sequence networks are used to study how the unbalanced instance affects the equivalent circuit. MATLAB/Simulink is used to evaluate and plot the transient and steady-state power fluctuations.

## 1 Objective

- To simulate a 440/33 $kV$, 112 kVA three phase transformer's single-phase lost (single line-to-ground/open phase) problem.
- To obtain transient and steady-state power fluctuations.
- To analyze the behaviour of positive-, negative-, and zero-sequence networks.

## 2 Transformer Specifications

- Rated Power: 112 kVA
- Primary Voltage: 440 V (L-L)
- Secondary Voltage: 33 kV (L-L)
- Frequency: 50 Hz
- Winding Connnection: $Y_g$-$Y_g$.

# 3 Theory

## 3.1 Balanced Operation

Under normal conditions, the three-phase circuit behaves in a balanced manner. In balanced condition, phasor summation of voltage and current should be zero and instantaneous power should be constant.

## 3.2 Unbalanced Operation (Phase Lost)

Under abnormal conditions, the three-phase circuit behaves unbalanced. In such a condition, phasor summation of voltage and current will not be zero and instantaneous power will not be constant, and its nature is oscillatory. Then we analyze the unbalanced condition, we use Fortescue theorem as follows:

- Positive-sequence network
- Negative-sequence network
- Zero-sequence network

## 3.3 Sequence Network Connection

The three sequence networks are as follows:

$$V_0 = \frac{V_a + V_b + V_c}{3} \tag{1}$$

$$V_1 = \frac{V_a + a^2 V_b + a V_c}{3} \tag{2}$$

$$V_2 = \frac{V_a + a V_b + a^2 V_c}{3} \tag{3}$$

where $a = e^{j120^\circ}$.

# 4 Simulink Implementation and Results

## 4.1 Steps

1. The 112kVA 440V/33kV three-phase transformer specifications are taken from [1] the parameters are $R_1 = R_2^{'} = 0.0086\Omega$, $L_1 = L_2^{'} = 0.1337mH$, $L_m = 0.224H$, $R_o = 17.416\Omega$. The parameters are taken on a per-phase basis referred to primary(LV).
2. Three ideal single phase transformers with a voltage supply per-phase of $254.03V$ are connected in a star connection, a 100 kW R-load is connected.
3. Voltage and current sensor blocks are used on all three phases.
4. A controlled breaker opens the secondary side of phase A at $t = 0.4$ s and remains open for the remainder of the simulation, $t = 1$ s.
5. During faults, the system becomes unbalanced and activates the positive-sequence, negative-sequence, and zero-sequence networks. We measure the secondary side voltage and power during fault condition.

6. Instantaneous power is computed using the following:

$$P(t) = v_a i_a + v_b i_b + v_c i_c \tag{4}$$

## 4.2 Simulink Diagram

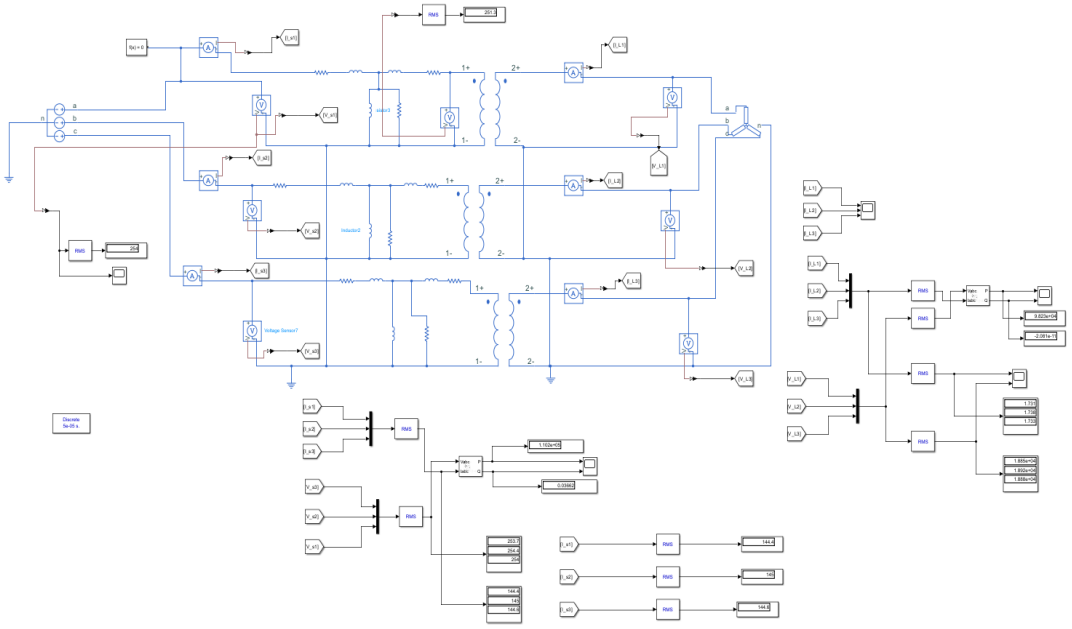


Figure 1: Simulink model of the transformer under balanced condition

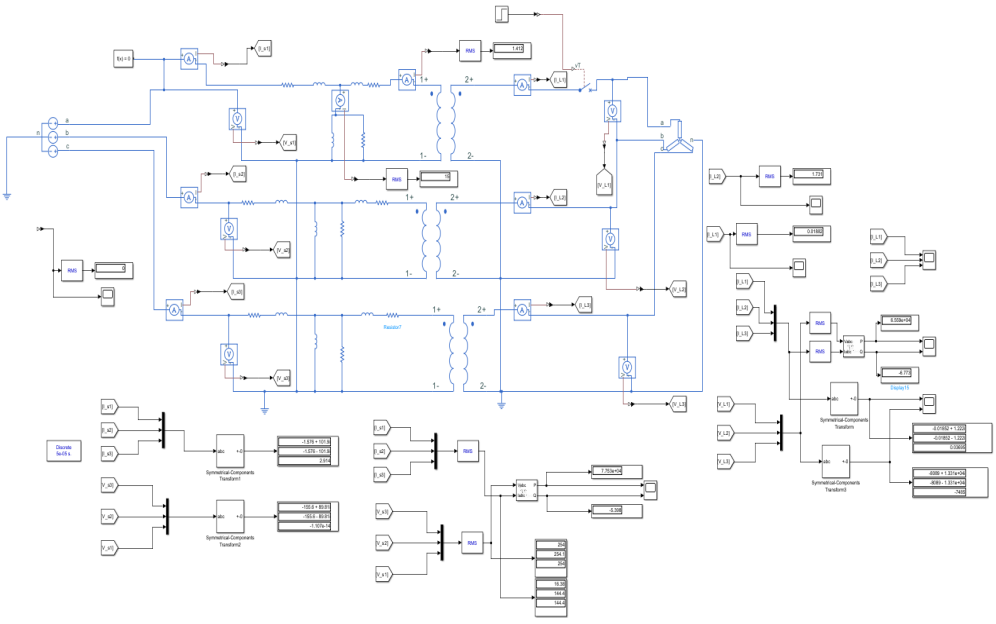


Figure 2: Simulink model of the transformer under unbalanced condition

# 5 Results

## 5.1 Phase Voltages

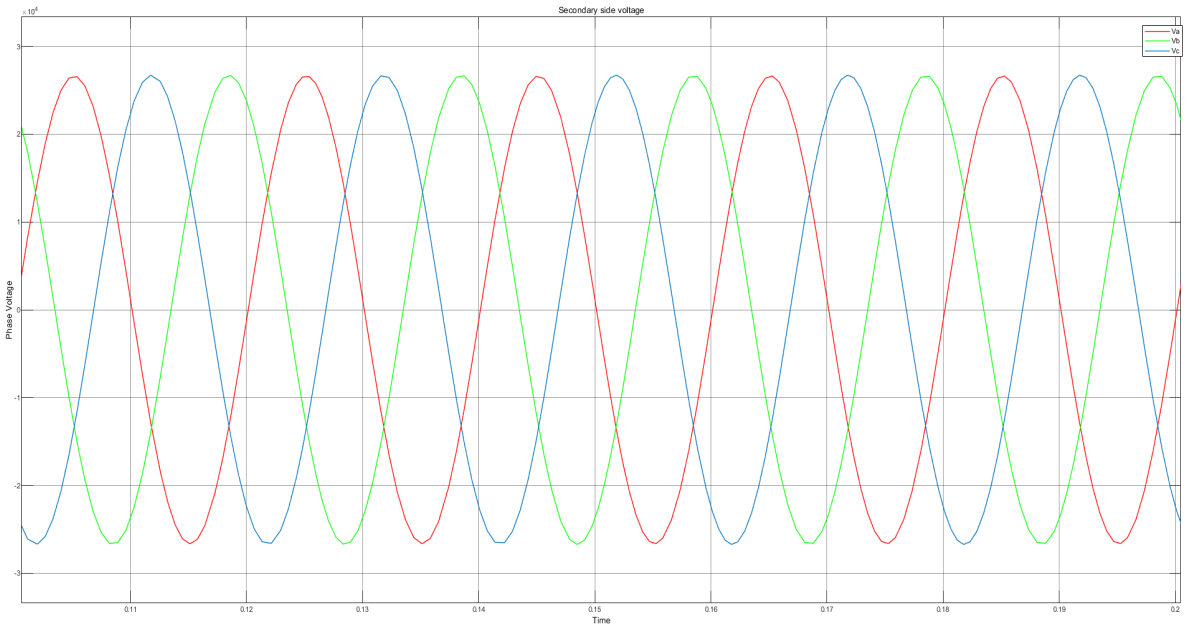



Figure 3: Phase voltages of secondary side showing balanced condition

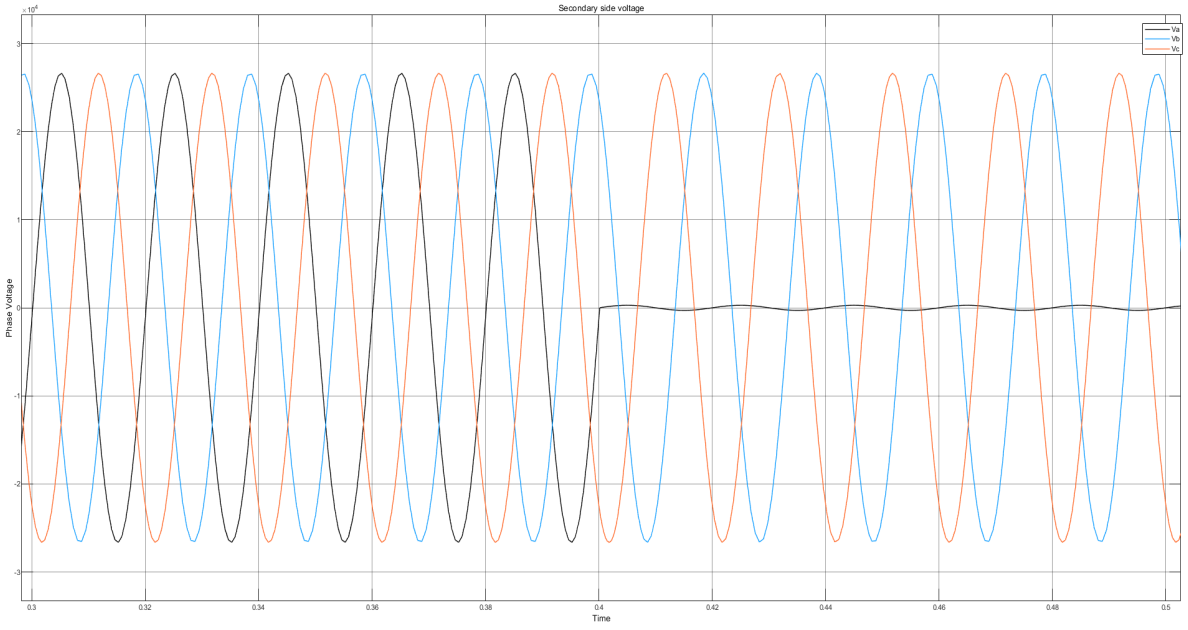



Figure 4: Phase voltages of secondary side showing unbalanced condition

## 5.2 Sequence Currents and Voltage

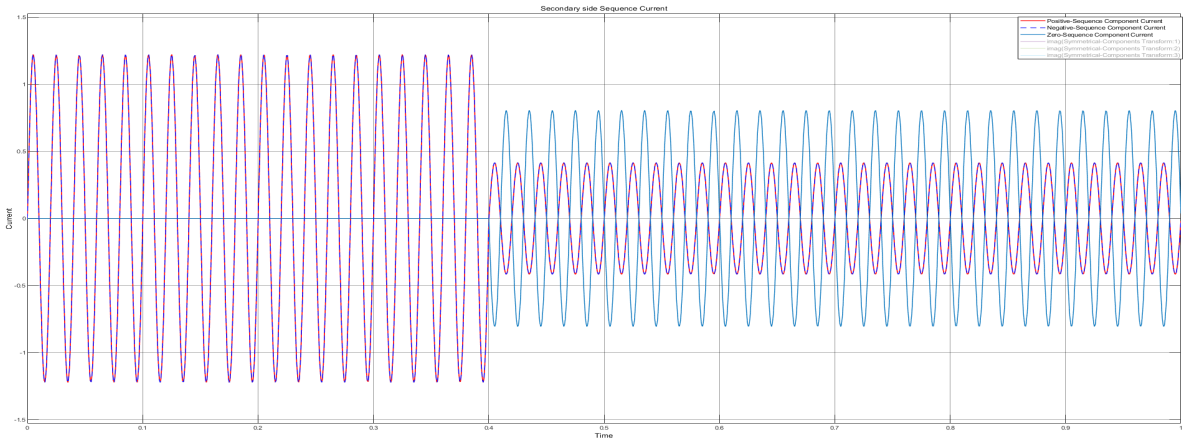



Figure 5: Positive, negative, and zero sequence voltage during fault

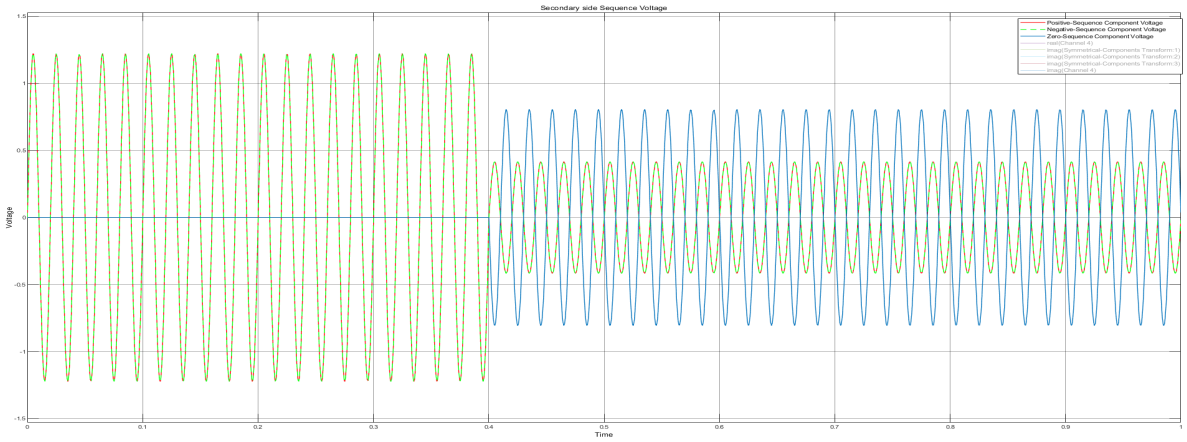



Figure 6: Positive, negative, and zero sequence voltage during fault

Table 1: Sequence components of voltage and current under abnormal condition

| Sequence Components | Voltage (V) | Current (A) |
|---|---|---|
| Positive Sequence component | 13313.937 $\angle 90.868^o$ | 1.222 $\angle 90.868^o$ |
| Negative Sequence component | 13313.937 $\angle -90.868^o$ | 1.222 $\angle -90.868^o$ |
| Zero Sequence component | 402.509 | 0.03696 |

### 5.3 Power Output Fluctuations

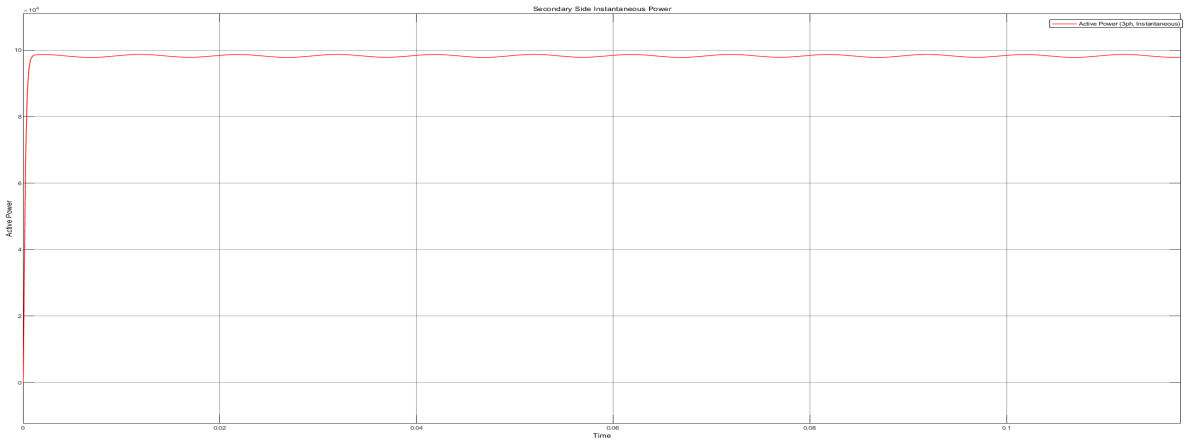



Figure 7: Instantaneous power under balanced condition

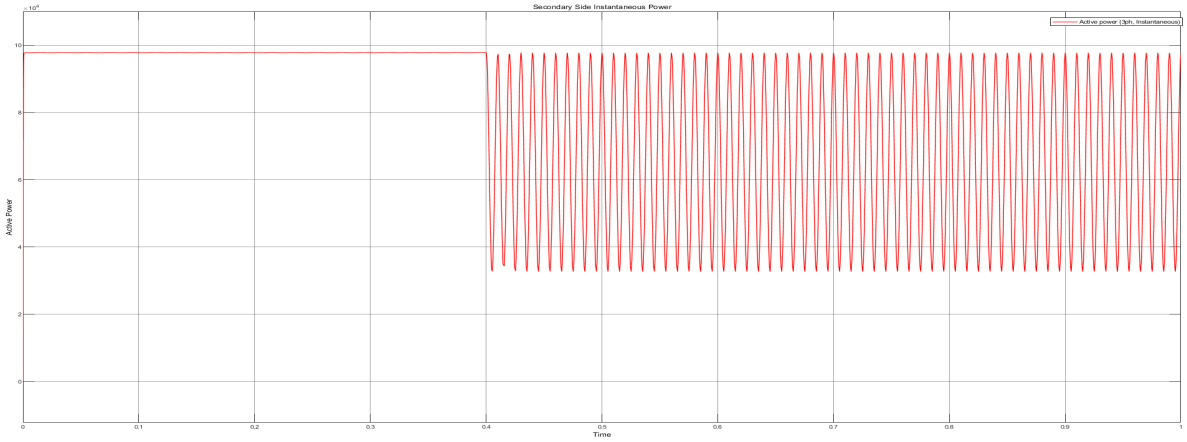



Figure 8: Transient and steady-state power instantaneous power under abnormal condition

## 6 Conclusion

The transformer shifts from a balanced to an unbalanced state due to a phase-lost fault, as the Simulink simulation effectively illustrates. The networks with positive, negative, and zero sequences start to function. Significant transients in power output are followed by a distorted steady-state profile.

## References

[1] CNC ELECTRIC GROUP CO., LTD, CNC Power transformer datasheet pg-34